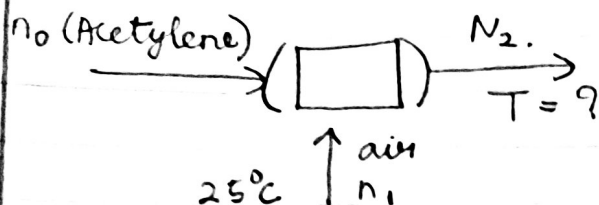


Homework 11

① adiabatically $Q = 0$

Acetylene:



$$n_{\text{O}_2, \text{in}} = n_0 \times \frac{5}{2} = \frac{5n_0}{2}$$

$$n_{\text{air, in}} = \frac{5n_0}{2} \times \frac{1}{0.21} = 11.905n_0$$

> Assume $n_0 = 1 \text{ mol}$

$$\Delta H_{\text{rxn}}^\circ = 4 \cdot -393.5 \text{ kJ/mol} + 2 \cdot -241.83 \text{ kJ/mol} - (2 \cdot 228.2 \text{ kJ/mol})$$

$$= -2514.06 \text{ kJ}$$

$$\Delta \hat{H}_{\text{rxn}}^\circ = \frac{\Delta H_{\text{rxn}}^\circ}{2 \text{ mol}} = -1257.03 \text{ kJ/mol}$$

$$Q = \sum \Delta \hat{H}_{\text{rxn}}^\circ + \sum n_e \hat{H}_e - \sum n_i \hat{H}_i^\circ = 0$$

$$n_{\text{C}_2\text{H}_2, \text{out}} = n_{\text{C}_2\text{H}_2, \text{in}} - \xi = 0$$

$$\xi = 1 \text{ mol}$$

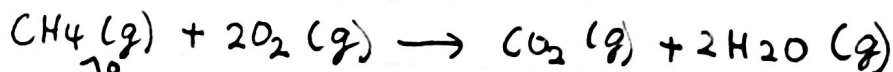
$$n_{\text{CO}_2, \text{out}} = n_{\text{CO}_2, \text{in}} + 2\xi = 2 \text{ mol}$$

$$n_{\text{H}_2\text{O, out}} = n_{\text{H}_2\text{O, in}} + 1\xi = 1 \text{ mol}$$

$$Q = 1 \text{ mol} \cdot -1257.03 \text{ kJ/mol} \times \frac{1000 \text{ J}}{1 \text{ kJ}} + 9.405 \text{ mol} \cdot 29.1 (x - 298.15 \text{ K}) + 2 \text{ mol} \cdot (x - 298.15 \text{ K}) + 1 \text{ mol} \cdot 33.6 (x - 298.15 \text{ K})$$

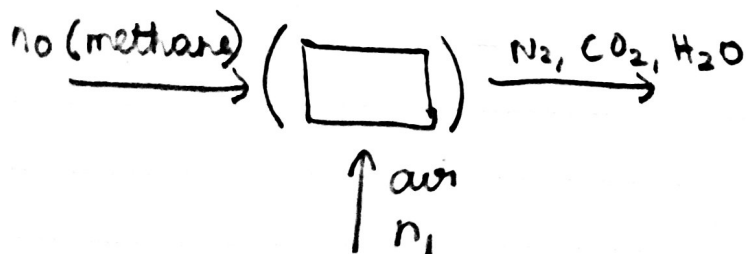
$$\Rightarrow x = 3594.97 \text{ K}$$

methane



$$\Delta \hat{H}_{\text{rxn}}^\circ = 1 \cdot -393.5 \text{ kJ/mol} + 2 \cdot -241.83 \text{ kJ/mol} - 1 \cdot 74.52 \text{ kJ/mol}$$

$$= -802.64 \text{ kJ/mol}$$



Assume $n_0 = 1 \text{ mol}$

$$n_{O_2, in} = 2 \text{ mol}$$

$$n_1 = 2 \times \frac{1}{0.21} = 9.524 \text{ mol}$$

$$n_{N_2, in} = n_{N_2, out} = 2 \times \frac{0.79}{0.21} = 7.524 \text{ mol}$$

$$n_{CH_4, out} = n_{CH_4, in} - \xi = 0$$

$$\xi = 1 \text{ mol}$$

$$n_{CO_2, out} = n_{CO_2, in} + \xi = 1 \text{ mol}$$

$$n_{H_2O, out} = n_{H_2O, in} + 2\xi = 2 \text{ mol}$$

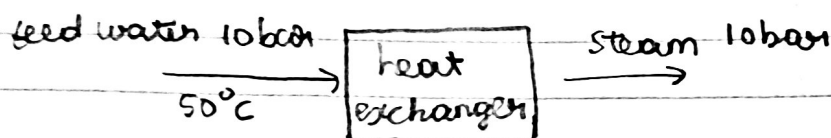
$$Q = \xi \Delta \hat{H}_{rxn} + \sum_e n_e \hat{H}_e - \sum_i n_i \hat{H}_i = 0$$

$$0 = 1 \text{ mol} \times -802.64 \text{ kJ/mol} \times \frac{1000 \text{ J}}{1 \text{ kJ}} + 7.524 \text{ mol} \cdot 29.1 (x - 298.15 \text{ K}) + 1 \text{ mol} \cdot 37 (x - 298.15 \text{ K}) + 2 \text{ mol} \cdot 33.6 (x - 298.15 \text{ K})$$

$$\Rightarrow x = 2843.86 \text{ K}$$

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(2)



$$X_1 = 0.4 \text{ (ethane)}$$

$$X_2 = 0.19 \text{ (N}_2\text{)}$$

$$n_1 X_3 = 0.01 \text{ (H}_2\text{S)}$$

$$X_4 = 0.1 \text{ (O}_2\text{)}$$

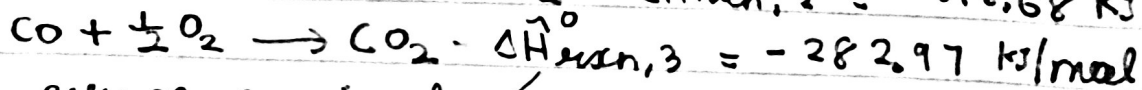
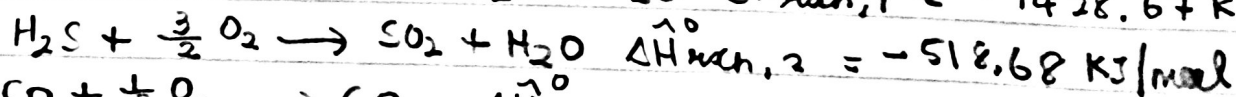
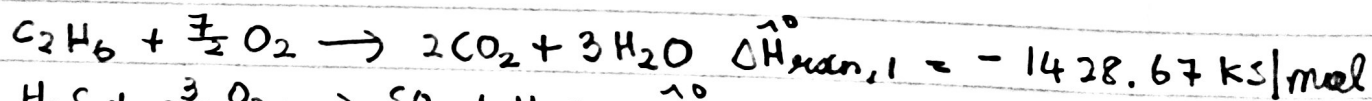
$$X_5 = 0.3 \text{ (CO)}$$

$$25^\circ\text{C}$$

$$O_2, N_2, CO_2, H_2O, SO_2$$

$$150^\circ\text{C}$$

$$n_2 \text{ (air)}$$



assume $n_1 = 1 \text{ mol}$

$$n_{\text{ethane}, out} = n_{\text{ethane}, in} - \xi_1 = 0$$

$$\xi_1 = 0.4 \text{ mol}$$

$$n_{H_2S, out} = n_{H_2S, in} - \xi_2 = 0$$

$$\xi_2 = 0.01 \text{ mol}$$

$$n_{CO, out} = n_{CO, in} - \xi_3 = 0$$

$$\xi_2 = 0.3 \text{ mol}$$

$$1.758$$

$$n_{O_2, required} = \frac{7}{2} \cdot \xi_1 + \frac{3}{2} \xi_2 + \frac{1}{2} \xi_3 = 1.565 \text{ mol}$$

Excess

$$n_{2, O_2} = 1.565 \text{ mol} \times 1.2 = 1.878 \text{ mol}$$

$$n_2 = 1.878 \text{ mol} \times \frac{1 - 0.21}{0.21} = 8.947 \text{ mol} \rightarrow 6.613$$

$$n_{O_2, out} = n_{2, O_2} - n_{O_2, required} + 0.1 n_1 = 0.413 \text{ mol}$$

$$n_{N_2, out} = n_2 \times 0.79 + 0.19 n_1 = 7.255 \text{ mol}$$

$$n_{CO_2, out} = n_{CO_2, in} + \xi_3 = 0.3 \text{ mol}$$

$$n_{SO_2, out} = n_{SO_2, in} + \xi_2 = 0.01 \text{ mol}$$

$$n_{H_2O, out} = n_{H_2O, in} + 3\xi_1 + \xi_2 = 1.21 \text{ mol}$$

$$Q = \xi_1 \Delta \hat{H}_{rxn, 1} + \xi_2 \Delta \hat{H}_{rxn, 2} + \xi_3 \Delta \hat{H}_{rxn, 3} + \sum n_e \hat{H}_e + \sum n_p \hat{H}_p$$

$$= (0.4 \text{ mol} \cdot -1428.67 \text{ kJ/mol} + 0.01 \text{ mol} \cdot -518.68 \text{ kJ/mol} + 0.3 \text{ mol} \cdot -282.97 \text{ kJ/mol})$$

$$+ 0.413 \text{ mol} \cdot 29.3 (423.15 - 298.15) + 7.255 \text{ mol} \cdot 29.1 (423.15 - 298.15)$$

$$+ 0.3 \text{ mol} \cdot 37 (423.15 - 298.15) + 0.01 \text{ mol} \cdot 39.9 (423.15 - 298.15)$$

$$+ 1.21 \text{ mol} \cdot 33.6 (423.15 - 298.15)$$

$$= -627124 \text{ J} \times \frac{1 \text{ kJ}}{1000 \text{ J}} = -627.124 \text{ kJ}$$

$$\hat{H}_{in} (10 \text{ bar}, 50^\circ\text{C}) = 210.19 \text{ kJ/kg}$$

$$\hat{H}_{out} (10 \text{ bar}, \text{saturated}) = 2777.1 \text{ kJ/kg}$$

$$Q = m (\hat{H}_{out} - \hat{H}_{in})$$

$$m = \frac{Q}{\hat{H}_{out} - \hat{H}_{in}} = \frac{627.124 \text{ kJ}}{(2777.1 - 210.197) \text{ kJ/kg}} = 0.244 \text{ kg/mol gas}$$

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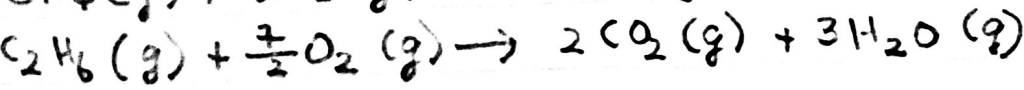
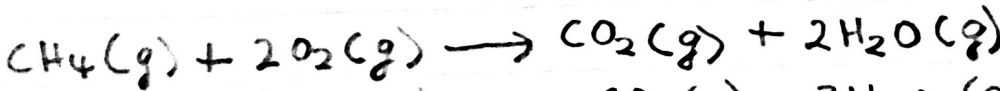
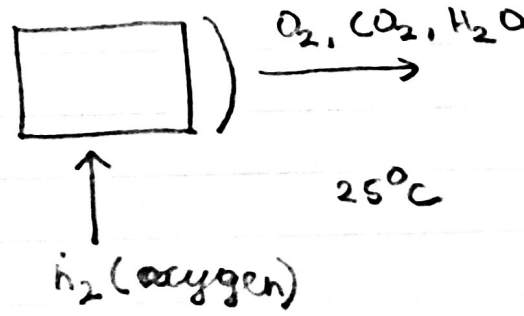
(3)

$$x_1 = 0.85 (\text{methane})$$

$$x_2 = 0.15 (\text{ethane})$$

$$n_1$$

$$25^\circ\text{C}$$



a)

$$\dot{n}_1 = 1 \text{ mol/s}$$

$$\dot{n}_{\text{meth}, \text{out}} = \dot{n}_{\text{meth}, \text{in}} - \dot{\xi}_1 = 0$$

$$\Rightarrow \dot{\xi}_1 = 0.85 \text{ mol/s}$$

$$\dot{n}_{\text{eth}, \text{out}} = \dot{n}_{\text{eth}, \text{in}} - \dot{\xi}_1 = 0$$

$$\dot{\xi}_1 = 0.15 \text{ mol/s}$$

$$\Delta \hat{H}_{\text{rxn}, 1}^{\circ} = 1 \cdot -393.5 \text{ kJ/mol} + 2 \cdot -241.83 \text{ kJ/mol} - 1 \cdot -74.52 \text{ kJ/mol} = -802.64 \frac{\text{kJ}}{\text{mol}}$$

$$\Delta \hat{H}_{\text{rxn}, 2}^{\circ} = (4 \cdot -393.5 \text{ kJ/mol} + 6 \cdot -241.83 \text{ kJ/mol} - 2 \cdot -83.82 \text{ kJ/mol}) / 2 = -1428.67 \text{ kJ/mol}$$

$$\dot{Q} = \dot{\xi}_1 \Delta \hat{H}_{\text{rxn}, 1}^{\circ} + \dot{\xi}_2 \Delta \hat{H}_{\text{rxn}, 2}^{\circ} + \sum \dot{n}_i \hat{H}_i^{\circ} - \sum \dot{n}_j \hat{H}_j^{\circ}$$

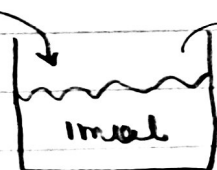
$$= 0.85 \text{ mol/s} \cdot (-802.64 \text{ kJ/mol}) + 0.15 \text{ mol/s} \cdot (-1428.67 \text{ kJ/mol})$$

$$= -896.5445 \text{ kJ/s} \quad \checkmark$$

$$\hat{Q} = \frac{\dot{Q}}{\dot{n}} = \frac{-896.5445 \text{ kJ/s}}{1 \text{ mol/s}} = -896.545 \text{ kJ/mol}$$

b)

O_2 25°C



$\rightarrow 20^\circ\text{C}, \text{O}_2, \text{CO}_2, \text{H}_2\text{O}$

$$\hat{Q} = (\dot{\xi}_1 \Delta \hat{H}_{\text{rxn}, 1}^{\circ} + \dot{\xi}_2 \Delta \hat{H}_{\text{rxn}, 2}^{\circ}) / 1 \text{ mol}$$

$$= -896.545 \text{ kJ/mol}$$

$$\dot{n}_{\text{meth}, \text{out}} = \dot{n}_{\text{meth}, \text{in}} - \dot{\xi}_1 = 0$$

$$\Rightarrow \dot{\xi}_1 = 0.85 \text{ mol}$$

$$\dot{n}_{\text{eth}, \text{out}} = \dot{n}_{\text{eth}, \text{in}} - \dot{\xi}_2 = 0$$

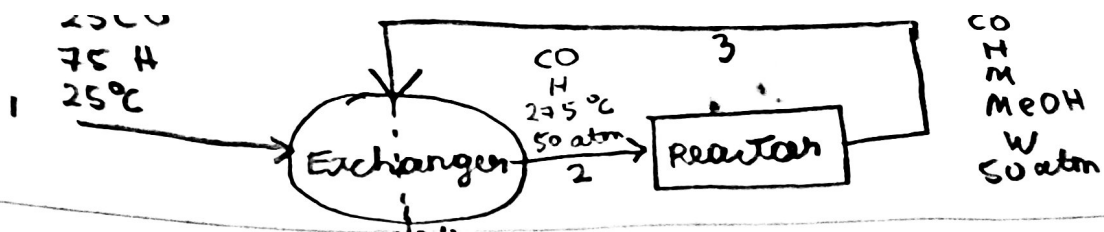
$$\dot{\xi}_2 = 0.15 \text{ mol}$$

It's Baten

$\hookrightarrow$ Internal E.

c)

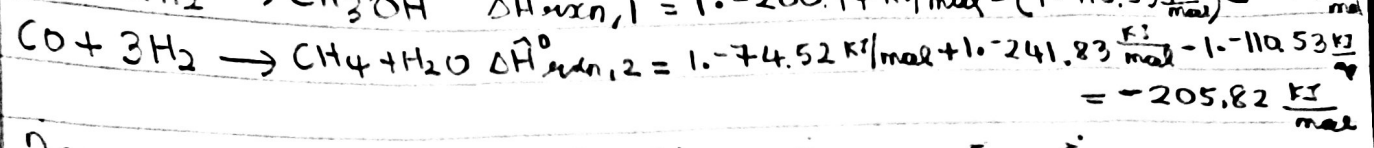
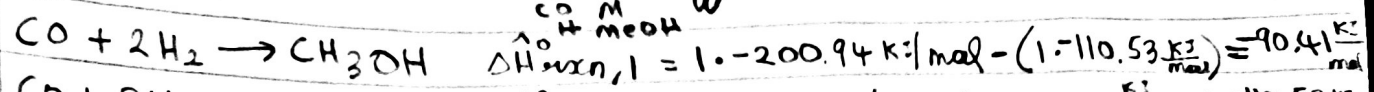
The results in parts (a) and (b) do not depend on the percent excess oxygen because since the temperature at inlet and outlet is both at 25°C, which is the reference temperature, so we can ignore them, so now $\dot{Q} = \dot{\xi}_1 \Delta \hat{H}_{\text{rxn}, 1}^{\circ} + \dot{\xi}_2 \Delta \hat{H}_{\text{rxn}, 2}^{\circ}$, and enthalpy of reaction excludes heat of formation of reactants and products, but O_2 is a pure element, so it has 0 heat of formation, which is the reason why answers for (a) and (b) do not depend on the percent excess oxygen.



4.

$$Q = 0$$

{



$$\dot{n}_{\text{CO},\text{out}} = 25 \text{ mol/s} \times 0.2 = 5 \text{ mol/s} = \dot{n}_{\text{CO},\text{in}} - \xi_1 - \xi_2$$

$$\dot{n}_{\text{CH}_3\text{OH},\text{out}} = \dot{n}_{\text{CH}_3\text{OH},\text{in}} + \xi_1 = 18.7 \text{ mol/s}$$

$$\xi_1 = 18.7 \text{ mol/s}$$

$$\xi_2 = 1.3 \text{ mol/s}$$

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$$\dot{n}_{\text{H}_2,\text{out}} = \dot{n}_{\text{H}_2,\text{in}} - 2\xi_1 - 3\xi_2 = 33.7 \text{ mol/s}$$

$$\dot{n}_{\text{CH}_4,\text{out}} = \dot{n}_{\text{CH}_4,\text{in}} + \xi_2 = 1.3 \text{ mol/s}$$

$$\dot{n}_{\text{H}_2\text{O},\text{out}} = \dot{n}_{\text{H}_2\text{O},\text{in}} + \xi_2 = 1.3 \text{ mol/s}$$

Differential Energy Balance: $\hat{H}_2 = \hat{H}_3$

To calculate the difference in enthalpy between streams 2 and 3, we construct a path:

- decrease the pressure of stream 2 from 50 atm to 1 atm
- decrease the temperature of stream 2 from 275°C to 25°C
- react streams 2 to stream 3 through reactions R1 and R2 at 25°C
- increase temperature of stream 3 from 25°C to T_{out}
- increase pressure of stream 3 from 1 atm to 50 atm

$$\left\{ \begin{aligned} C_{p,\text{CO}} &= 30.87 - 0.01285T + 2.729 \times 10^{-5}T^2 - 1.272 \times 10^{-8}T^3 \\ C_{p,\text{H}} &= 27.14 + 0.0093T - 1.381 \times 10^{-5}T^2 + 7.645 \times 10^{-9}T^3 \\ C_{p,\text{MeOH}} &= 21.15 + 0.07092T + 2.587 \times 10^{-5}T^2 - 2.852 \times 10^{-8}T^3 \\ C_{p,\text{CH}_4} &= 19.25 + 0.05213T + 1.177 \times 10^{-5}T^2 - 1.132 \times 10^{-8}T^3 \\ C_{p,\text{H}_2\text{O}} &= 32.24 - 0.001924T + 1.055 \times 10^{-5}T^2 - 3.596 \times 10^{-9}T^3 \end{aligned} \right.$$

Excel sheet T_3

$$\hat{H}_3 - \hat{H}_2 = \sum_c \dot{n}_{c,2} \int_{275}^{25} C_{p,i} dT + \sum_k \xi_k \Delta \hat{H}_{\text{rxn},k} + \sum_c \dot{n}_{c,3} \int_{25}^{T_3} C_{p,i} dT = 0$$

The solver analytical tool was used to solve for T_3 . From solver, we get $T_3 = 1300 \text{ K} \approx 1020^\circ\text{C}$ (reactor effluent T)

$$\hat{H}_2 - \hat{H}_1 = \hat{H}_3 - \hat{H}_4$$

Energy balance equation would therefore be:

$$\sum_i n_{i2} \int_{25}^{275} c_{pi} dT = \sum_i n_{i3} \int_{T_4}^{1000} c_{pi} dT$$

By using values, we get $T_4 = 1060 \text{ K} \cong 790^\circ\text{C}$
 (Product outlet Temp)

Therefore, the temperature of the reactor effluent drops by 230°C